

Around or Across? Radius, Diameter, Area, Perimeter

Directions:

- Before computing, decide two things: is the question asking for the distance **around** (perimeter, circumference) or the surface **covered** (area)? And if a circle is involved, were you given the **radius** or the **diameter**?
- Round answers with π to two decimal places and include the unit on the answer line.
- Two problems show another student's work with a mistake. Find it, then fix it.

1. A circular clock face has a radius of 6 cm. Find its circumference.

Answer: _____ cm

2. A round pizza has a *diameter* of 10 inches. Find its area.

Answer: _____ in²

3. A rectangular garden measures 9 m by 4 m. How many meters of fence are needed to enclose it?

Answer: _____ m

4. A rectangular rug is 7 ft long and 5 ft wide. How much floor does it cover?

Answer: _____ ft^2

5. A circular pond has a *diameter* of 14 m. Leo walks once around its edge. How far does he walk?

Answer: _____ m

6. Mara finds the area of a circle whose *diameter* is 16 cm. Her work:

$$A = \pi r^2 = \pi \cdot 16^2 = 804.25 \text{ square centimeters}$$

(a) Name the error in Mara's work.

(b) Find the correct area.

Answer: _____ cm^2

7. Jon wants enough paint to cover a rectangular wall stripe 12 ft long and 3 ft tall. His work:

$$2 \cdot (12 + 3) = 30, \text{ so he buys paint for 30 square feet}$$

(a) What went wrong? Explain in one sentence.

(b) Find the amount of wall the paint must actually cover.

Answer: _____ ft^2

8. Challenge. A circular table top has a radius of 0.6 m. A metal trim runs around its edge, and a glass sheet covers its top.

(a) How long is the trim?

(b) How large is the glass sheet?

Answer: _____ m

Answer: _____ m^2